

Yu Han

Research Associate | University of Cambridge

Urban and regional planning · Sustainable mobility · Climate resilience

Email: yh593@cam.ac.uk | Mobile: +44-07349331830

ORCID: 0000-0003-2361-9802 | LinkedIn: [yu-han-79998865](#)

Website: <https://yuh2017.github.io/yuhan/>

Research profile

My research examines the interactions between land use, transport systems and socioeconomic change, with a focus on sustainable urban development and climate resilience. I integrate quantitative spatial analysis, machine learning, agent-based simulation and urban digital twins to assess how development patterns and planning interventions shape mobility, accessibility and climate risk.

Research areas: Land use and transport planning; sustainable mobility; climate resilience and adaptation; urban digital twins and AI; spatial equity and social vulnerability.

Education

PhD in Urban and Regional Planning	2016–2020
University of Florida	
Dissertation: Agent-based Modeling of Human-Environment Interactions in Coastal Flood Adaptation	
MSc in Electrical and Computer Engineering	2018–2019
University of Florida	
MSc in Civil and Environmental Engineering	2012–2014
Carnegie Mellon University	
BSc in Environmental Engineering	2008–2012
Beijing Institute of Technology	

Academic appointments

Research Associate	Dec 2024–present
University of Cambridge, UK Department of Land Economy	
Research Assistant Professor	Jun 2023–Jun 2024
University of Twente, the Netherlands Faculty of Geo-information Science and Earth Observation (ITC)	
Lecturer	Aug 2022–May 2023
Capital University of Economics and Business, China Urban Economics and Public Administration	
Postdoctoral Associate	Oct 2021–Apr 2022
Texas A&M University, College Station, USA Landscape Architecture and Urban Planning	
Postdoctoral Fellow	Dec 2020–Oct 2021
Florida International University, USA Department of Earth and Environment	
Research Assistant	Aug 2016–Dec 2020
University of Florida, USA Urban and Regional Planning	
Research Fellow	Oct 2014–Aug 2016
Tsinghua University, China Institute of Environment	

Research projects and contributions

University of Cambridge | Research Associate | 2024–present

DARe project: national scenarios for an adaptable and resilient transport system

Funding: UK Research and Innovation and UK Department for Transport

Integrated land use and agent-based transport simulation framework for climate resilience and adaptation.

- Model and project national commuting flows using graph neural networks, linking residential and employment patterns with alternative socioeconomic scenarios.
- Integrate commuting flows with National Travel Survey data to generate population profiles and daily travel activities for agent-based transport simulation.
- Combine data from multiple sources, simulation and scenario analysis in urban digital twin modeling to assess climate-hazard exposure and support adaptation planning.
- Prepare national socioeconomic and mobility datasets; assess road transport resilience and visualise future scenarios.

University of Twente | Assistant Professor | 2023–2024

PARATUS: indirect impacts of compound hazards on transport systems

Funding: EU Horizon programme

- Developed methodologies using open data to assess indirect impacts of compound and cascading hazards on transport systems.
- Analysed cascading impacts through impact chains and contributed to a web-based stakeholder support tool for systemic disaster risk assessment.

Texas A&M University | Postdoctoral Associate | 2021–2022

Human-centred urban digital twins for coastal communities

Funding: National Science Foundation

- Integrated infrastructure and social datasets to support planning simulation and coastal adaptation analysis.
- Combined modelling perspectives and visualisation of adaptation scenarios within digital twin research.

Florida International University | Postdoctoral Fellow | 2020–2021

Decentralised resilience in interdependent infrastructure systems

Funding: National Science Foundation | Role: participant

- Developed a risk-based model of long-term storm surge damage, using cloud computing and multithreaded programming to analyse large numbers of coastal buildings.
- Incorporated social vulnerability and public adaptation policies into scenario analysis; evaluated uncertainty through Monte Carlo simulation and global sensitivity analysis.

University of Florida | Research Assistant | 2019–2020

Life Cycle Costs and Benefits Analysis of Freight Transportation Projects

Funding: Florida Department of Transportation | Role: Co-PI

- Developed life-cycle cost methods using FHWA guidance and a Python interface for cost–benefit analysis; evaluated transport investments through FDOT case studies, reports and presentations.

Teaching and graduate supervision

Global Challenge & Local Actions | Project Tutor | Spring 2024

University of Twente

- Supported group projects connecting global environmental change with local coastal planning challenges and quantitative analysis.

Modeling Multi-Hazard & Risks | Instructor | Fall 2023

University of Twente

- Combined computational exercises with hazard and risk case studies, helping students interpret modelling outputs for planning decisions.

Introduction of Urban Data Science | Co-teaching | Fall 2022

Capital University of Economics and Business

- Taught Python-based spatial analytics and data-driven approaches to urban planning.

URP6821 Transportation and Land Use Modeling | Teaching Assistant | Spring 2020

University of Florida

- Supported seven laboratory sessions on metropolitan transport planning and the four-step travel modelling process; helped students apply theory in course projects.

Graduate supervision and student support

- Supervised graduate thesis research at the University of Twente.
- Teaching approach emphasises critical thinking, structured demonstrations, hands-on analysis and progressively independent project work.

Research proposal development

- Focused CoPe: Climate Change, Coastal Hazards, and Human Health (C3H3) Hub. National Science Foundation Coastlines and People (CoPe), 2021. PI: Pallab Mozumder.
- Enhancing the Understanding of Multi-(Hazard-)Risk by Identifying Synergies between Climate Change Adaptation and Disaster Risk Reduction. VU-UT Alliance, 2023.
- Systemic Risk Assessment for Compounding and Cascading Events in Disaster Risk Reduction and Climate Adaptation. VU-UT Alliance, 2024.

Quantitative and technical skills

Methods: Agent-based modelling; land use and transportation planning; scenario analysis; risk analysis; life-cycle cost–benefit assessment.

Programming: Python, R, Java, MATLAB, C and C++; Linux.

Spatial and simulation tools: GIS, MATSim, PyTorch.

Databases: PostGIS and MySQL.

Document preparation: LaTeX.

Languages

English: fluent speaking, reading and writing.

Chinese: native speaking, reading and writing.

Published articles

1. Xinmeng Shan, **Yu Han**, Jiahong Wen, Hengzhi Hu, Jiao Wang, Guotao Zhang, Paolo Scussolini, Qian Ke, Mengya Li, Jun Wang (*). (2025) **Integrating Nature-based Solutions for Compound Flood Risk Mitigation in China: A Case Study of Shanghai**. *Journal of Environmental Management*.
<https://doi.org/10.1016/j.jenvman.2025.125155>
2. Changjie Chen (*), **Yu Han**, Andrea Galinski, Christian Calle, Jeffery Carney, Xinyue Ye, Cees van Westen. (2025) **Integrating urban digital twins with cloud-based geospatial dashboards for coastal resilience planning: A case study in Florida**. *Journal of Planning Education and Research*.
<https://doi.org/10.1177/0739456X251316185>
3. Zan Wang, Shengwen Qi(*), **Yu Han**, Bowen Zheng, Yu Zou, Yue Yang.(2025) **Delimitation of Landslide Areas in Optical Remote Sensing Images across Regions via Deep Transfer Learning**. *IEEE Access*.
[doi:10.1109/ACCESS.2024.3514216](https://doi.org/10.1109/ACCESS.2024.3514216)
4. **Yu Han** (*), Wei Zhai, Pallab Mozumder, Cees van Westen, Changjie Chen. (2025) **Modeling evacuation activities amid compound hazards: insights from the case of Hurricane Irma in southeast Florida**. *Travel Behaviour and Society*. 38, 100933. <https://doi.org/10.1016/j.tbs.2024.100933>
5. **Yu Han**, Xinyue Ye (*), Chunwu Zhu. (2024) **The Unequal Impact of Disasters: Assessing the Interplay Between Social Vulnerability, Public Assistance, Flood Insurance, and Migration in the US**. *Urban Informatics*. 3, 30. <https://doi.org/10.1007/s44212-024-00061-9>
6. Wei Zhai, Mengyang Liu, **Yu Han**. (2024) **Dynamic neighborhood isolation and resilience during the pandemic in America's 50 largest cities**. *Cities*. 153, 105260. <https://doi.org/10.1016/j.cities.2024.105260>
7. **Yu Han** (*), Xinyue Ye, Kayode Atoba, Pallab Mozumder, Changjie Chen, Bastian van den Bout, Cees van Westen. (2024). **Retreat from flood zones: Simulating land use changes in response to compound flood risk in coastal communities**. *Cities*, 149, 104953. <https://doi.org/10.1016/j.cities.2024.104953>. (★)
8. **Yu Han** (*), Haifeng Jia, Changqing Xu, Marija Bockarjova, Cees van Westen, Luigi Lombardo. (2023). **Unveiling spatial inequalities: Exploring county-level disaster damages and social vulnerability on public disaster assistance in the contiguous US**. *Journal of Environmental Management*, 351, 119690.
<https://www.sciencedirect.com/science/article/pii/S0301479723024787>
9. Xinyue Ye (*), Jiaxin Du, **Yu Han**, Galen D. Newman, David Retchless, Lei Zou, Youngjib Ham, Zhenhang Cai, **Developing Human-Centered Urban Digital Twins for Community Infrastructure Resilience: A Research Agenda**. *Journal of Planning Literature*. <https://doi.org/10.1177/08854122221137861>
10. **Yu Han**, Xinyue Ye (*) (2022). **Examining the effects of flood damage, federal hazard mitigation assistance, and flood insurance policy on population migration in the conterminous US between 2010 and 2019**. *Urban Climate*. <https://doi.org/10.1016/j.uclim.2022.101321>
11. **Yu Han**, Xiao Huang, Xinyue Ye(*), Bahar Dadashova(2022). **Adaptation planning and hazard mitigation for interdependent infrastructure systems to enhance urban resilience under climate change**. *Landscape Architecture Frontiers*. [DOI:10.15302/J-LAF-1-030031](https://doi.org/10.15302/J-LAF-1-030031)
12. Zhengsong Lin, Yuting Wang, Xinyue Ye, Yuxi Wan, Tianjun Lu, **Yu Han** (2022). **Effects of Low-Carbon Visualizations in Landscape Design Based on Virtual Eye-Movement Behavior Preference**. *Land*.
<https://www.mdpi.com/2073-445X/11/6/782>

Published articles continued

13. Changqing Xu, Xinmei Shi, Mingyi Jia, **Yu Han**, Rongrong Zhang, Shakeel Ahmad, Haifeng Jia(*) (2022), **China Sponge City database development and urban runoff source control facility configuration comparison between China and the US**, *Journal of Environmental Management*, <https://doi.org/10.1016/j.jenvman.2021.114241>
14. **Yu Han**(*), Liang Mao, Xuqi Chen, Wei Zhai, Zhong-Ren Peng, Pallab Mozumder (2022), **An agent-based model to simulate human-environment interactions in community flood adaptation**. *Risk Analysis*. <https://onlinelibrary.wiley.com/doi/10.1111/risa.13854> (✱)
15. **Yu Han** (*), Pallab Mozumder (2022), **Risk-based flood adaptation assessment for coastal buildings based on cloud computing**. *Sustainable Cities and Society*, <https://doi.org/10.1016/j.scs.2021.103415>
16. **Yu Han** (*), Pallab Mozumder (2021), **Building-level Adaptation Analysis under uncertain Sea-Level Rise**. *Climate Risk Management*, <https://doi.org/10.1016/j.crm.2021.100305>
17. **Yu Han**, Changjie Chen, Zhong-Ren Peng (*), Pallab Mozumder (2021), **Evaluating impacts of coastal flooding on the transportation system using an activity-based travel demand model: a case study in Miami-Dade County, FL**. *Transportation*, <https://doi.org/10.1007/s11116-021-10172-w>. (✱)
18. **Yu Han**, Kevin Ash, Liang Mao, Zhong-Ren Peng (*) (2020). **An agent-based model for community flood adaptation under uncertain sea-level rise**. *Climatic Change*, <https://link.springer.com/article/10.1007/s10584-020-02802-6>
19. **Yu Han**, Zhong-Ren Peng (*) (2019) **The integration of local government, residents, and insurance in coastal adaptation: An agent-based modeling approach**. *Computers, Environment and Urban Systems*, <https://www.sciencedirect.com/science/article/abs/pii/S0198971518305362?via%3Dihub>. (✱)
20. Wei Zhai, Xueyin Bai, Yu Shi, **Yu Han**, Zhong-Ren Peng (*), Chaolin Gu (2019). **Beyond Word2vec: An approach for urban functional region extraction and identification by combining Place2vec and POIs**. *Computers, Environment and Urban Systems*, <https://doi.org/10.1016/j.compenvurbsys.2018.11.008>.
21. **Yu Han**, Haifeng Jia (*) (2017). **Simulating the spatial dynamics of urban growth with an integrated modeling approach: A case study of Foshan, China**. *Ecological Modelling*, <https://www.sciencedirect.com/science/article/abs/pii/S0304380016301028?via%3Dihub>

Working papers

1. Xu Zhang, **Yu Han** (*), Shaokun Guo, Jie Dong, Ling Chang, InSAR-informed coastal flood risk assessment under sea-level rise in Miami-Dade County, Florida. GIScience & Remote Sensing (Under Review)
2. **Yu Han**, Ismail Saadi, Phil Blythe, Li Wan, Integrating Gravity-Informed Graph Neural Networks into Agent-Based Simulation for Scenario Planning: A Case Study in Cambridgeshire and Peterborough, UK. (Draft Completed; Manuscript in Preparation)
3. **Yu Han**, Ismail Saadi, Li Wan, Developing National Adaptation Scenarios to Model Commuting Flows Across the UK Using Gravity-Based Graph Neural Network Models. (Draft Completed; Manuscript in Preparation)
4. **Yu Han**, Changjie Chen, Jeff Carney, Li Wan, A Human-Centered Urban Digital Twin for Climate-Hazard Exposure Assessment and Adaptation Planning. (Under Preparation)

Professional report and developed software

Professional report

Zhong-Ren Peng, Yu Han. Life Cycle Costs and Benefits Analysis of Freight Transportation Projects, BDV31 977-119. Florida Department of Transportation (FDOT).

<https://rosap.ntl.bts.gov/view/dot/56929>

Developed software

- Yu Han, Zhong-Ren Peng. Life Cycle Cost and Benefit Analysis of Freight Transportation, FreightEC 2.0. Florida Department of Transportation (FDOT), 2020.
- Yu Han, Haifeng Jia, Weize Song. A land use change simulation system based on multi-criteria evaluation. Tsinghua University, 2015.

Conferences

Yu Han, Cees van Western. 2024. Modeling Evacuation Strategies in Response to Compound Hazards: Lessons Learned from a Major Hurricane Event in the US. EGU General Assembly 2024, Vienna, Austria, 14–19 Apr 2024, EGU24-9975, <https://doi.org/10.5194/egusphere-egu24-9975>, 2024.

Cees Van Westen, Bastian van den Bout, Rabina Twayana, Massimiliano Pittore, Ashok Dahal, Manzul Hazarika, **Yu Han**. A web-based multi-hazard risk simulation service based on impact chains, EGU General Assembly 2024, Vienna, Austria, 14–19 Apr 2024, EGU24-15139, <https://doi.org/10.5194/egusphere-egu24-15139>, 2024.

Wang Zan, Shengwen Qi, Youshan Liu, Bowen Zheng, Peng Sun, **Yu Han**. Unsupervised Deep Learning for Rapid Subsurface Interface Identification using Geophysical Measurements, EGU General Assembly 2024, Vienna, Austria, 14–19 Apr 2024, EGU24-19107, <https://doi.org/10.5194/egusphere-egu24-19107>, 2024.

Yu Han, Xinyue Ye. Symposium on Human Dynamics Research: Geospatial modeling of climate risk and community resilience. AAG 2022 Annual Meeting (virtual), <https://aag-annualmeeting.secure-platform.com/a/solicitations/19/sessiongallery/2893>, 2022

Conference presentations continued

Yu Han, Xinyue Ye. Agent-based modeling to evaluate community flood adaptation strategies using cloud computing. AAG 2022 Annual Meeting (virtual), <https://aag-annualmeeting.secure-platform.com/a/solicitations/19/sessiongallery/2893/application/8136>, 2022

Yu Han. 2021. Examining Social and Institutional Barriers on Flood Resilience in the US. ACSP 2021 Annual Conference (virtual)

Yu Han, Changjie Chen, Ruth L. Steiner, Zhong-Ren Peng. Evaluating Impacts of Coastal Flooding on the Transportation System Using Activity-based Modeling: a Case Study in Miami-Dade County, FL. Proceedings of the Transportation Research Board (TRB) 100th Annual Meeting, Washington, D.C., USA, <https://trid.trb.org/view/1918925>, 2021

Yu Han. 2020. Probabilistic Assessment of Community Flood Adaptation based on Cost-Benefit Analysis under Sea-Level Rise. Virtual SRA 2020 Annual Meeting (Presentation video available: https://www.youtube.com/watch?v=wmfGi6H2Y_Q).

Yu Han, Zhong-Ren Peng. 2019. Agent-based Modeling of Individual Behavior Dynamics on Flood Disaster Risk Assessment: A Case Study in Miami Beach. The 2018 ACSP 58th Annual Conference, Buffalo, New York

Yu Han, Zhong-Ren Peng. 2019. Evaluating the Accessibility of Highway Network and Community Cost from the Damage of Coastal Storms. Proceedings of the Transportation Research Board (TRB) 98th Annual Meeting, Washington, D.C., USA

Yu Han, Zhong-Ren Peng. 2018. Agent-based Modeling of the Evolution of Attitudes on Flooding Risk in Coastal Areas in Miami. The 2017 American Association of Geographers (AAG) annual meeting, New Orleans, Louisiana

Academic service

Reviewer for international journals and conferences

Journal of Planning Education and Research; Travel Behaviour & Society; Natural Hazards Review; Scientific Reports; Climate Resilience and Sustainability; Sustainable Cities & Society; Computers, Environment and Urban Systems; Ecological Modelling; Nature Climate Change; Nature Sustainability; Transportation Research Board (TRB).

Awards and scholarships

- 2016–2020 | University of Florida Graduate Research Assistant Scholarship.
- 2020 | Paul and Malea Zwick Graduate Student Award, Urban and Regional Planning, University of Florida (\$550).
- 2019 | Institute of Transportation Engineers, University of Florida Travel Award (\$700), TRB Annual Meeting, Washington, DC.
- 2018 | Urban and Regional Planning, University of Florida Travel Award (\$200), ACSP 58th Annual Conference, Buffalo, New York.
- 2017 | Urban and Regional Planning, University of Florida Travel Award (\$400), AAG Annual Meeting, New Orleans, Louisiana.
- 2012–2013 | Carnegie Mellon University Partial Graduate Tuition Waiver.
- 2009–2010 | Beijing Institute of Technology University Undergraduate Scholarship.